Response to reviewer comments on "Generalized Jeffreys's approximate objective Bayes factor: Model-selection consistency, finite-sample accuracy, and the Bayesian evidence in early-phase clinical trials"

Puneet Velidi, Zhengxiao Wei, Evan Strasdin, Shreena Nisha Kalaria, Yimeng Liu, Céline M. Laumont, Brad H. Nelson, Farouk S. Nathoo

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Response to Associate Editor's Comments

The paper introduces a generalized form of Jeffreys's approximate objective Bayes factor (eJAB). It was read by myself and two expert referees. The authors establish consistency and assess finite-sample accuracy via simulations. There were numerous major concerns identified by the reviewers:

1. *'more exploration of the connection to test-based BF (Johnson, 2005)'*

   We originally selected test-statistic Bayes factors (TSBFs) for two main reasons: their computational convenience, as they admit closed-form expressions (although TSBFs are not 'approximate' Bayes factors in the same sense as eJAB), and the lack of readily available MCMC-based Bayes factors for chi-squared and other nonparametric tests considered in our simulations. In our response to Reviewer 1's comment, we now discuss the assumptions and asymptotic properties of three versions of the TSBF and compare them with those of eJAB. We have also revised the simulation studies to avoid comparisons between eJAB and the minimum Bayes factor version of TSBF.

2. *'significant concerns about the regularity condition R2'*

   In our response to Reviewer 1, we have carefully addressed concerns regarding the regularity condition R2 required for model-selection consistency. First, we have discussed asymptotic properties of TSBF in relation to R2 in our first response to Reviewer 1. In our second response to Reviewer 1, we have made comparisons with the BIC approximation to the Bayes factor and the conditions required for model selection consistency in relation R2. Finally, we have conducted an empirical investigation of R2 based on data obtained from four major replication studies, where we observe realizations of D_n under H_1 and comment on the behavior of D_n with increasing sample size.

3. *'significant concerns about the analysis in Section 4.'*

   We have revised the analysis in response to Reviewer 1's suggestions. The application is now restricted to 16,563 primary- and secondary-outcome analyses from 1,790 early-phase trials, with detailed inclusion

and exclusion criteria provided in Supplementary Material Appendix C. One-sided tests are now considered explicitly in addition to two-sided tests. In the revision we have defined a one-sided version of eJAB and discussed its asymptotic properties.

4. *Addition of new coauthor:* We have added one new coauthor to the manuscript. Evan Strasdin is a graduate student who conducted an empirical study that forms the second part of our response to the second comment of Reviewer 1. We have added this study as a new section in the Supplementary Material as Appendix E.

5. *Minor modification to the title of the manuscript:* We have changed the title of the manuscript from '*Generalized Jeffreys's approximate objective Bayes factor: Model-selection consistency, finite-sample accuracy, and statistical evidence in 71,126 clinical trial findings*' to '*Generalized Jeffreys's approximate objective Bayes factor: Model-selection consistency, finite-sample accuracy, and the Bayesian evidence in early-phase clinical trials*'. This change is required as part of our response to the fifth comment of Reviewer 1, who suggested that we only look at early phase trials (phase I and II) in our analysis.

Response to Reviewer 1

- *The proposed BF depends on the test-statistic of a chi-squared test with dimension q. It also depends on sample size n. The connection to the test-based BF (Johnson, 2005) should be explored in more detail, both theoretically and through simulation. For example, Johnson also considers BFs based on the F- (rather than the chi-squared) statistic, which also depend on sample size. In fact, you do use F-tests in simulation items 4 and 5. A closer theoretical comparison with BFs based on a chi-squared test statistic is warranted. The derivation in Johnson is based on certain asymptotic assumptions (local alternatives); how do these compare with assumptions R1 and, in particular, R2?*

Thank you for raising this question. We have expanded the discussion to distinguish three related constructions: the local alternative prior, the maximum marginal likelihood estimate (MMLE), and the nonlocal alternative prior. Johnson's TSBF models the sampling distribution of a classical test statistic rather than the full-data likelihood. For the chi-squared test, the statistic has a central chi-squared distribution under $\mathcal{H}_0$ and a noncentral chi-squared distribution under $\mathcal{H}_1$, yielding a closed-form Bayes factor that depends on the statistic x_n, its degrees of freedom q, the sample size n, and an alternative-model dispersion parameter $\varphi = (c+1)/n$, where $c = \mathcal{O}(n)$ denotes the Dirichlet concentration parameter,

$$\mathcal{H}_0 : \; x_n \sim \chi^2_q, \tag{1}$$

$$\mathcal{H}_1 : \; x_n \mid \gamma \sim \chi^2_q(\gamma), \qquad \gamma \mid \varphi \sim \text{Gamma}\left(\frac{q}{2}, \frac{\varphi}{2}\right), \tag{2}$$

$$TSBF_{10} = \left(\frac{\varphi}{\varphi+1}\right)^{\frac{q}{2}} \exp\left\{\frac{x_n}{2(\varphi+1)}\right\}. \tag{3}$$

Johnson (2005) considers maximizing the marginal density of the observed test statistic under $\mathcal{H}_1$ over the dispersion parameter. This produces the minimum BF_{01} within the specified class (Held and Ott, 2018). It is therefore an upper bound and the least conservative assessment of evidence against $\mathcal{H}_0$. Our previous manuscript compared eJAB only with this MMLE version of TSBF, rather than with the broader class of TSBFs (Johnson, 2005, 2008; Yuan and Johnson, 2008; Johnson, 2013). Because this quantity is maximized in favor of $\mathcal{H}_1$, it is bounded below by one and equals one when the observed statistic does not exceed its expectation under $\mathcal{H}_0$, producing the characteristic 'hockey-stick' behavior. We agree that comparing an approximate Bayes factor with a minimum Bayes factor is an apples to oranges comparison. Accordingly, we have removed the comparisons with the MMLE-TSBF and replaced them with comparisons to the TSBF more generally.

The asymptotic assumptions differ in their intent. To test the hypotheses $\mathcal{H}_0 : \boldsymbol{\theta} = \boldsymbol{\theta}_0$ versus $\mathcal{H}_1 : \boldsymbol{\theta} \neq \boldsymbol{\theta}_0$, Johnson's original chi-squared derivation assumes local alternatives satisfying $\boldsymbol{\theta} - \boldsymbol{\theta}_0 = \mathcal{O}_p(n^{-\frac{1}{2}})$. In other words, a local alternative not only centers a unimodal, symmetric prior distribution at the null value but also specifies that the prior scale shrinks with the sample size. One consequence of this assumption is the existence of a nondegenerate asymptotic distribution for the test statistic under $\mathcal{H}_1$—specifically, a noncentral chi-squared limit with an $\mathcal{O}_p(1)$ noncentrality parameter (Johnson, 2005, Lemma 1). It is not a condition that ensures model-selection consistency. Under this local-alternative regime, regularity condition R2 will not hold, and model-selection consistency of eJAB is not guaranteed. However, we note that the TSBF is also not model-selection consistent under the same assumption. Under $\mathcal{H}_0$, $x_n \xrightarrow{D} \chi^2_q$ and $\varphi \to \varphi_0 > 0$ as $n \to \infty$, so $TSBF_{10}$ in Eq. 3 is nondegenerate and almost surely positive. Thus, under Johnson's local-alternative framework, the asymptotic TSBF exists in a nondegenerate form, but neither eJAB nor the TSBF is model-selection consistent. By contrast, eJAB is based on three asymptotic assumptions: (1) Asymptotic normality of the posterior distribution; (2) R1: $p \xrightarrow{D} \text{Unif}(0,1)$ as $n \to \infty$ under $\mathcal{H}_0$; (3) R2: $D_n = -\sqrt{n} \cdot p \ln p \xrightarrow{P} 0$ as $n \to \infty$ under $\mathcal{H}_1$. The first assumption along with the assumed prior gives us the form for eJAB. The latter two assumptions are sufficient for model-selection consistency (i.e., Theorem 2.3). Under these assumptions, the same argument shows that $TSBF_{10}$ in Eq. 3 does not converge to zero under $\mathcal{H}_0$ and remains

model-selection inconsistent. We have revised the manuscript to clarify the differences in assumptions and their implications.

We further consider the model selection consistency of newer versions of TSBF developed in subsequent work, across three recent papers, the Bayes factor function (BFF) retains the same core structure—a family of TSBFs evaluated over alternative hypotheses indexed by standardized effect size ω—while the principal extensions concern the nonlocal priors used to define those alternatives, expanding the first-order moment and gamma priors to arbitrary-order moment and inverse-moment/inverse-gamma families and examining how their shape and dispersion affect evidence accumulation, operating characteristics, and aggregation across replicated studies (Johnson et al., 2023; Datta et al., 2025a,b). The following expressions give the BFF for the chi-squared test proposed by Johnson et al. (2023),

$$\mathcal{H}_0 : \; x_n \sim \chi^2_q, \tag{1}$$

$$\mathcal{H}_1 : \; x_n \mid \gamma \sim \chi^2_q(\gamma), \qquad \gamma \mid \tau^2 \sim \text{Gamma}\left(\frac{q}{2}+1, \, \frac{1}{2\tau^2}\right), \tag{4}$$

$$BFF_{10}\left(x_n \mid \tau^2\right) = (1+\tau^2)^{-\frac{q}{2}-1}\left(1+\frac{\tau^2 x_n}{q(1+\tau^2)}\right)\exp\left\{\frac{\tau^2 x_n}{2(1+\tau^2)}\right\}, \tag{5}$$

$$\tau^2 = \frac{n}{q}\omega^2. \tag{6}$$

They assume the alternative in Eq. 4 so that the prior density on the noncentrality parameter vanishes at the origin. Since $\text{mode}(\gamma) = q\tau^2$, Eq. 6 matches the prior mode to the noncentrality $n\omega^2$ corresponding to a specified standardized effect size ω. This first-order nonlocal prior yields faster accumulation of evidence in favor of a true null than the corresponding local prior. Under $\mathcal{H}_0$, $x_n = \mathcal{O}_p(1)$ and, for every fixed $\omega > 0$, $\tau^2 = \mathcal{O}(n)$, so $BFF_{10} = \mathcal{O}_p\left(n^{-\frac{q}{2}-1}\right) \xrightarrow{P} 0$ as $n \to \infty$. The faster polynomial accumulation of null evidence motivates the higher-order nonlocal priors considered in Datta et al. (2025a). Under a fixed alternative with true standardized effect $\omega_0 > 0$, the true noncentrality parameter satisfies $\gamma = n\omega_0^2$, and hence $x_n = n\omega_0^2 + \mathcal{O}_p(\sqrt{n})$. For any fixed nonzero BFF ordinate $\omega > 0$, $\tau^2 = \mathcal{O}(n)$, and therefore the exponential term in Eq. 5 dominates its polynomial penalty. In particular, $\frac{1}{n}\ln BFF_{10} \xrightarrow{P} \frac{\omega_0^2}{2}$, so $BFF_{10} \xrightarrow{P} \infty$ as $n \to \infty$. Thus, for every fixed nonzero effect-size ordinate ω, the BFF is pointwise model-selection consistent.

- *More theoretical motivation and discussion are needed for regularity condition R2. It is a stronger condition than assuming that the traditional minimum BF calibration, $-e \cdot p \ln p$ (Sellke et al., 2001), converges to zero under $\mathcal{H}_1$ as n goes to infinity. So how strong is condition 2? Is this assumption justifiable?*

Assume that, under $\mathcal{H}_1$, the p-value is bounded away from 1 for all sufficiently large n. Then, $-e \cdot p \ln p \xrightarrow{P} 0$ is equivalent to $p \xrightarrow{P} 0$ because $\lim_{x \to 0^+} x \ln x = 0$. Thus, requiring the traditional minimum-BF calibration to converge to zero imposes no restriction on the rate at which the p-value converges to zero under $\mathcal{H}_1$. With regard to the strength of assumption R2: $D_n = -\sqrt{n} \cdot p \ln p \xrightarrow{P} 0$ as $n \to \infty$ under $\mathcal{H}_1$, we first note that it is not sufficiently strong to ensure model-selection consistency of the BIC approximation to the Bayes factor, as formulated in Eq. 7. Let $Q_{\chi^2_q}(1-p)$ denote the quantile function of a chi-squared distribution with q degrees of freedom, evaluated at tail probability $1-p$, where $p \to 0^+$. Suppose that the p-value is computed from a likelihood-ratio test so that, under standard regularity conditions,

$$BF_{01}^{(\text{BIC})} = n^{\frac{q}{2}} \exp\left\{-\frac{1}{2}Q_{\chi^2_q}(1-p)\right\}, \tag{7}$$

and that R2 holds, but $-n^a \cdot p \ln p \xrightarrow{P} 0$ for $a > \frac{1}{2}$ as $n \to \infty$ under $\mathcal{H}_1$. Then, $\exists \epsilon > 0$ such that $\mathbb{P}(-n^a \cdot p \ln p < \epsilon) \nrightarrow 1$, which, in turn, implies that $\mathbb{P}(BF_{01}^{(\mathrm{BIC})} < \epsilon) \nrightarrow 1$ for $q > 1$ under $\mathcal{H}_1$.

More specifically, using the asymptotic expansion of the incomplete gamma function (e.g., Daouia et al., 2025, Lemma A.2), $\frac{1}{2} Q_{\chi^2_q}(1-p) = -\ln p + (\frac{q}{2} - 1)\ln(-\ln p) - \ln\Gamma(\frac{q}{2}) + o(1)$. It follows that $BF_{01}^{(\mathrm{BIC})} \approx \Gamma(\frac{q}{2})\, n^{\frac{q}{2}} \cdot p\,(-\ln p)^{1-\frac{q}{2}}$, which requires a substantially stronger condition than R2 for model-selection consistency when $q > 1$.

We next consider condition R2 from a more practical standpoint. We conduct a study examining the realized values of D_n in four large-scale replication projects. Each project selected a collection of previously published findings and attempted to replicate them under a preregistered protocol, so that each original result was paired with an independently conducted replication of the same effect.

- **EERP**, the Experimental Economics Replication Project (Camerer et al., 2016), which attempted to replicate laboratory experiments published in the *American Economic Review* and the *Quarterly Journal of Economics* between 2011 and 2014;
- **RPP**, the Reproducibility Project: Psychology (Open Science Collaboration, 2015), which attempted to replicate a collection of findings drawn from three high-impact psychology journals;
- **RPCB**, the Reproducibility Project: Cancer Biology (Errington et al., 2014), which considered experiments reported in high-impact preclinical cancer biology papers; we use the project's final analysis (Errington et al., 2021); and
- **SCORE**, Systematizing Confidence in Open Research and Evidence (Tyner et al., 2025), which assembled and considered replication of a large corpus of published claims from the social and behavioral sciences.

For each result in the four projects, we extracted the original p-value and sample size n, and the p-value and sample size of each of its replication attempts. Each result has at least one and possibly several replication attempts. Our objective is to examine realized values of D_n under H_1 as a function of sample size. To do so, we create an $\mathcal{H}_1$ subset corresponding to replicated results where it is safe to assume that $\mathcal{H}_1$ is true. A result is included in the $\mathcal{H}_1$ subset when its original p-value satisfies $p \leq 0.005$ and all of its replication attempts also reached $p \leq 0.005$. Given that the effect was found at this level in the original study and in every replication, we make the assumption that $\mathcal{H}_1$ is true for these results. This gives realized values of D_n under $\mathcal{H}_1$: one from each original study and one from each replication attempt. Table 1 reports the counts of the results reported and used in our study.

Two kinds of recorded p-value were treated separately. Where a p-value was recorded as 0, we computed $\ln p$ from the reported test statistic and, where the test has them, its degrees of freedom. Of the plotted values, 51 were obtained this way, including the value at the largest sample size; 12 values (six SCORE original results and six SCORE replication attempts) could not be computed and were dropped. Where a p-value was reported as an upper bound, such as $p < .001$, rather than as a value, we removed it. Such left-censored p-values occur only in SCORE. SCORE labels its original p-values; we removed the 37 it labels as less-than or threshold values, and the one with no label. SCORE does not label replication p-values, so we treated a replication p-value recorded as an exact power of ten, such as .001, as censored unless we were able to recompute the p-value directly from the reported test statistic. This removed two replication values where the p-value is left censored and cannot be recalculated from the reported test statistic.

Figure 1 plots the realized values of D_n against the sample size n, which spans roughly seven orders of magnitude. A clear pattern emerges indicating that the distribution of D_n is shifted more toward zero as n increases. The value of D_n at the largest value of n, $n = 49{,}196{,}731$ with $D_n = 4.09 \times 10^{-29}$, is circled in the figure. Overall, the study suggests that R2 is practically feasible.

The two largest sample sizes depicted in Figure 1 are one SCORE claim and its replication. The sample sizes are unusually large, so we provide some information about these studies. The original study from 2014 (Travers and Krezmien, 2018) examined autism identification among US public school students, using counts of students with autism from the IDEA Data Center and enrolment from the NCES Digest of Education Statistics. The claim SCORE tested is that, compared with California, Ohio has lower odds of autism identification, assessed by a logistic regression of autism identification on state, with each student as one observation. The replication repeated this analysis with the same two sources for 2017, on 49,196,731 students in 46 states; SCORE records 7,977,660 students for the original. These

Project	Field	Results				(5) Plotted values	
		(1) released	(2) with p, n	(3) $p \leq 0.005$	(4) $\mathcal{H}_1$ subset	original	replication
EERP	Experimental economics	18	18	7	5	5	5
RPP	Psychology	167	143	67	16	16	16
RPCB	Preclinical cancer biology	158	115	61	15	15	21
SCORE	Social and behavioral sciences	331	311	184	102	58	107
	Total	674	587	319	138	94	149

Table 1: Counts at each step of the selection, by project. Column 1 *Released*: the number of results in each project's released data. A result is a study for EERP (18) and RPP (167, of which 100 were replicated), an effect for RPCB (158 effects from 23 papers, given in 188 rows), and a claim for SCORE (331 claims with at least one replication report). Column 2 *With p and n*: the results in Column 1 for which both the original p-value and the original sample size are recorded. Column 3 *Original $p \leq 0.005$*: the results in Column 2 whose original p-value satisfies $p \leq 0.005$. Column 4 $\mathcal{H}_1$ *subset*: the results in Column 3 for which every replication attempt reached $p \leq 0.005$. A result with no recorded replication p-value is not included; this excludes 20 RPP results and one RPCB result. We also excluded RPP study 69. The RPP data record $p = 10^{-6}$ for both its original and its replication, with no test statistic from which either value could be checked, whereas the reported p-values in the same data are $p < .05$ for the original and $p = .05$ for the replication. For SCORE, a replication attempt is one replication report. Columns 5 and 6 *Plotted values*: the number of values of D_n shown in Figure 1, one for each original result and one for each replication attempt in the $\mathcal{H}_1$ subset. This excludes left censored p-values that cannot be recomputed from the reported test statistics. For RPCB and SCORE the replication column exceeds the $\mathcal{H}_1$ subset because a result can have more than one replication attempt (up to three in RPCB and four in SCORE), each contributing its own value.

sample sizes are counts of enrolled students taken from administrative records. The estimated effects are small, with odds ratios of 0.935 in the original and 0.912 in the replication ($z = -6.31$ and $z = -12.32$).

This study is now reported in Appendix E of the Supplementary Material, and the data, code and counts for Table 1 and Figure 1 are in the GitHub repository. We refer to it at the end of Section 3.1 of the revision.

- *The analysis described in Section 4 is unconvincing for several reasons. First, clinical trial findings are usually based on one-sided tests at significance level $\alpha = 0.025$.*

  In the revised analysis, we determine the test direction from the reported statistical method. For explicitly one-sided scalar tests ($q = 1$), we apply the one-sided eJAB discussed below; for two-sided or otherwise unrestricted tests, we apply the two-sided eJAB. This classification yields 427 one-sided and 16,136 two-sided or unrestricted results in the revised analysis. When the test direction is not reported, we assign the analysis to the latter category. Given the reported numbers of one-sided and two-sided tests, this assumption seems reasonable. Supplementary Material Appendix C.5 provides the classification rules and describes the treatment of reported results, including censored p-values. Figure 3 combines one-sided and two-sided/unrestricted tests. A contradiction requires eJAB > 1 (one-sided or two-sided as appropriate) and $p \leq .05$ for two-sided tests or $p_1 \leq .025$ for one-sided tests. Across the clinical plots, we use the common comparison value $p_{\text{NHST}} = p$ for two-sided tests and $p_{\text{NHST}} = 2p_1$ for one-sided tests; the sweep requires $p_{\text{NHST}} \leq \alpha$. The folded value $2\min\{p_1, 1 - p_1\}$ reconstructs the unsigned statistic inside the one-sided eJAB formula and is not used for NHST rejection. A large p_1 does not reject in the specified direction, so it cannot yield a Bayes/NHST contradiction merely because $1 - p_1$ is small.

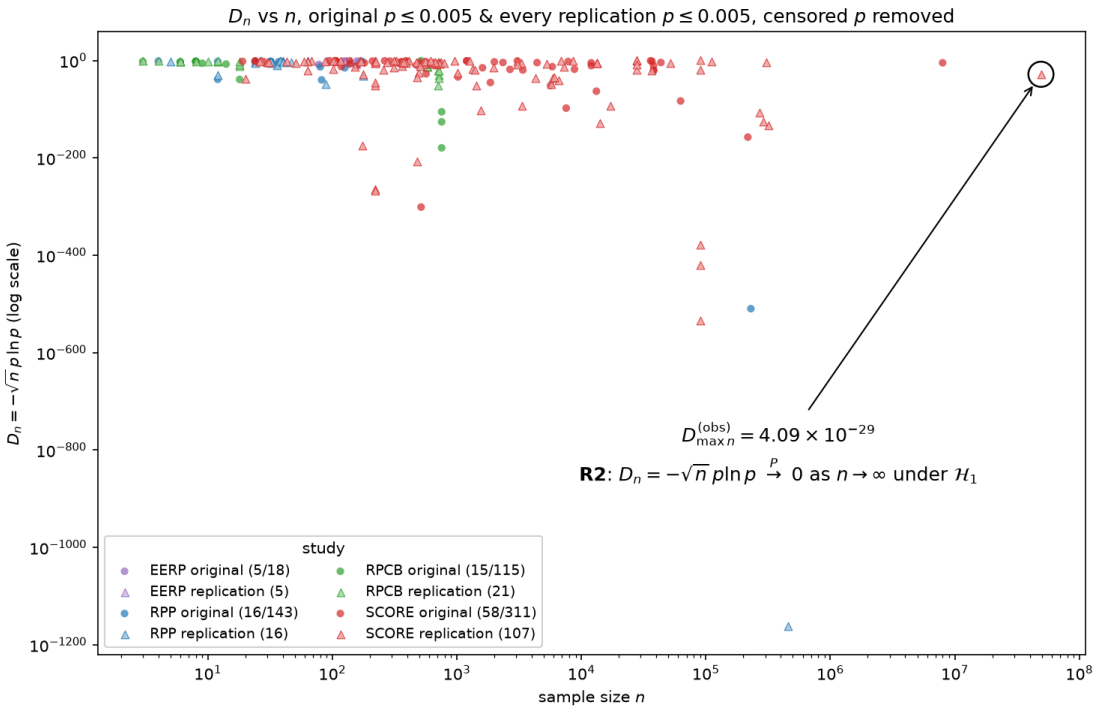


Figure 1: Realized values of $D_n = -\sqrt{n} \cdot p \ln p$, on a logarithmic scale, plotted against sample size n for the $\mathcal{H}_1$ subsets of the four replication projects (original $p \leq 0.005$ and every replication attempt $p \leq 0.005$). Circles denote original results and triangles replication attempts, with symbols colored by project. Legend entries give the number of plotted values; for original results this is shown relative to column 2 of Table 1. p-values reported as bounds are excluded. The circled value, $D_{\max n}^{(\text{obs})} = 4.09 \times 10^{-29}$, is the observed D_n at the largest sample size.

- *The proposed BF, however, is intrinsically two-sided, as it is based on a chi-squared statistic. Can you amend eJAB to the one-sided setting, taking into account the direction of the effect? A one-sided Bayes factor may need different priors and assumptions from those used here.*

  We have extended eJAB to the one-sided setting, e.g., $\mathcal{H}_0 : \theta = \theta_0$ versus $\mathcal{H}_+ : \theta > \theta_0$, and added the corresponding discussion, which explicitly incorporates the direction of the effect and the assumptions required for the one-sided Bayes factor. We have accordingly provided Appendix C.5 in the Supplementary Material.

  The encompassing-prior correction of Wagenmakers (2022, pp. 20–22) multiplies the Bayes factor for the null against an unrestricted alternative by the ratio of prior to posterior probability assigned to the specified direction. For a scalar right-sided test, we use a symmetric local prior centered at θ_0, assigning prior mass $1/2$ to the right-sided alternative and approximating the corresponding posterior mass by $1 - p_1$. With $p = 2\min\{p_1, 1 - p_1\}$, combining the null-centered Savage–Dickey approximation with the directional correction yields

$$eJAB_{0+} = \frac{\sqrt{n}}{2(1-p_1)} \exp\left\{-\frac{1}{2} Q_{\chi^2_1}(1-p)\right\}.$$

  When $q = 1$, the two $eJAB$ definitions are related by

$$eJAB_{0+} = \frac{n^{\frac{1}{2(1-n)}}}{2(1-p_1)} eJAB_{01}^{\frac{n}{n-1}}.$$

Under $\mathcal{H}_0$, **R1** implies $eJAB_{01} \xrightarrow{P} \infty$. Since $n^{\frac{1}{2(1-n)}} \to 1$ and $(2(1-p_1))^{-1} \geqslant 1/2$, we have $eJAB_{0+} \xrightarrow{P} \infty$. Under $\mathcal{H}_+$, **R2** implies $eJAB_{01} \xrightarrow{P} 0$. Assuming additionally **R3**: $(1-p_1)^{-1} = \mathcal{O}_p(1)$ under $\mathcal{H}_+$, it follows that $eJAB_{0+} \xrightarrow{P} 0$. Therefore, the one-sided $eJAB$ is also model-selection consistent.

- *Second, if you are concerned that 'failures at later phases may reflect type I errors, where the null hypothesis was incorrectly rejected at an earlier phase', then you should investigate only early-phase studies (Phases I and II) in your analysis. However, the analysis includes Phase II and even Phase IV studies.*

  Puneet: This response requires some resolution. First the raw proportion of NHST significant results where eJAB >1 is too far away from the posterior densities to make sense. Why? Are the raw proportions using eJAB > 1 while the psterior densities are based on eJAB > 1/3? Also, in the revision the new data the analysis has become less interesting. Even after resolving the discrepancy, can we make this more interesting? For example, find some, say 5 or so, good example of JLP where p is borderline and eJAB is large (say, eJAB >3). Then document them in a table that has a column describing the finding. Having some focus on JLP (examples where eJAB is large and p is borderline will hopefully make the analysis more interesting.) To address this concern, we now restrict the manuscript primary- and secondary-outcome analysis to 16,563 eligible analyses from 1,790 exact Early Phase I, Phase I, and Phase II interventional trials. Combined-phase trials and Phase III and IV trials are now excluded. We report the Bayes/NHST overall contradiction-rate profiles and posterior densities of the probability of contradiction among statistically significant NHST results separately for the three classifications of clinical trial.

  For trial phase g, let N_g count all eligible analyses, $S_g(\alpha)$ those meeting the NHST rule ($p \leq \alpha$ for two-sided tests or $p_1 \leq \alpha/2$ for one-sided tests), and $C_g(\alpha)$ those *also* having applicable $\text{eJAB}_{01} > 1$. Figure 2 plots $C_g(\alpha)/N_g$: the overall fraction of contradictions at level α. We note that this rate increases approximately linearly with the type I error rate α. Figure 3 shows the posterior densities of the overall proportion of NHST-significant results at $\alpha = .05$ that show evidence in favor of the null according to the eJAB Bayes factor approximation. The corresponding point estimate is $C_g(.05)/S_g(.05)$. These observed proportions are 6/94 (6.38%) in Early Phase I, 28/927 (3.02%) in Phase I, and 459/5,453 (8.42%) in Phase II. Their denominators combine one-sided results with $p_1 \leq .025$ and two-sided/unrestricted results with $p \leq .05$. Of the 6,474 significant results, 135 are one-sided; none of those 135 has eJAB > 1.

  Figure 3 quantifies how often conventional statistical significance coexists with evidence favoring the null and compares this conditional proportion across trial phases. The posterior densities are overlaid on common axes, with means of approximately 24.1%, 16.3%, and 18.1% for Early Phase I, Phase I, and Phase II, respectively. These indicate appreciable model-estimated shares of NHST-significant results favoring the null. The estimated proportion decreases from Early Phase I to Phase I and increases in Phase II, with greater uncertainty in Early Phase I and no monotonic phase pattern. The next sentence is a concern. It seems strange to me that the raw proportions are so different. Why? I have a hard time accepting that the prior causes this - please double check. As it is, I don't think we can report these results because they are too far away from the raw proportions. Is eJAB > 1.3 being used for the posterior results? If so, then lets use that for both the raw proportions and posterior results. These estimates differ from the raw observed proportions because the Dirichlet$(1, 1, 1, 1)$ prior adds one pseudocount to each of four categories in every study, with a substantial effect when study counts are sparse. The phase comparisons are descriptive.

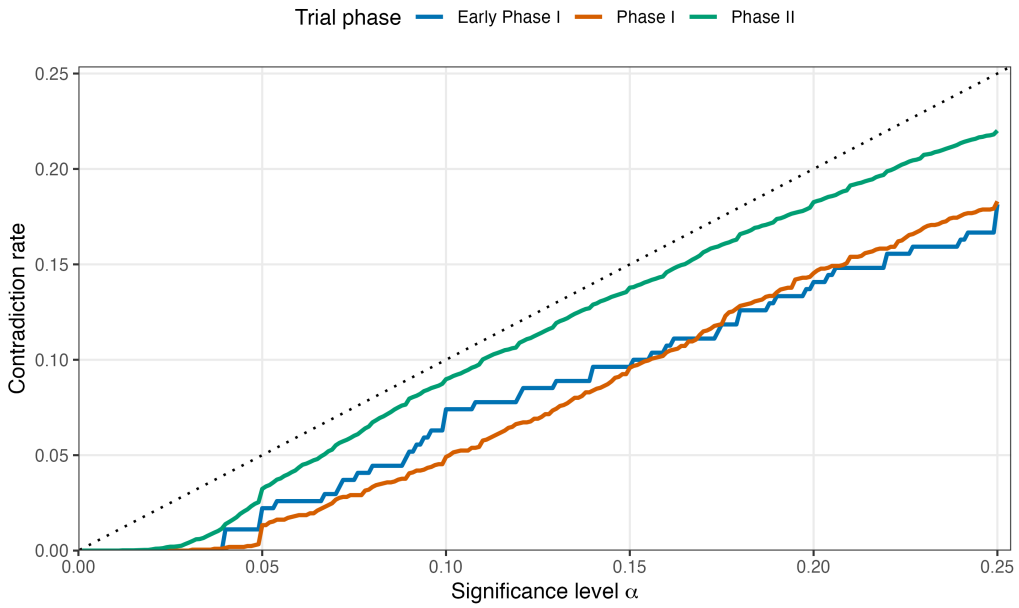


Figure 2: Overall Bayes/NHST contradiction proportion $C_g(\alpha)/N_g$ versus nominal significance level α, combining both test types within each phase g. The numerator counts results with $p \leq \alpha$ (two-sided) or $2p_1 \leq \alpha$ (one-sided), together with the applicable eJAB > 1; the denominator is all eligible analyses. At $\alpha = .05$, this uses the same contradiction count as the raw conditional proportion underlying Figure 3, but divides by all results rather than only significant results.

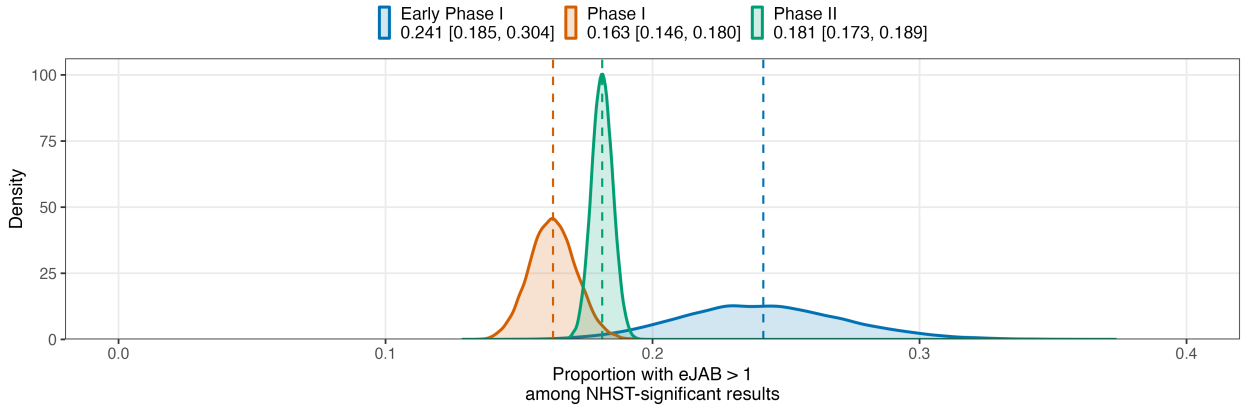


Figure 3: Overlaid posterior densities of the conditional contradiction proportion: the share of NHST-significant results with applicable eJAB > 1, combining one-sided tests ($p_1 \leq .025$) and two-sided/unrestricted tests ($p \leq .05$). The raw counterpart is $C_g(.05)/S_g(.05)$, whereas Figure 2 uses all eligible results in the denominator. Common axes allow comparison across phases; legend entries give posterior means [95% credible intervals], and dashed lines mark means. Study-level category probabilities are weighted by eligible-analysis counts before conditioning on significance. The per-study Dirichlet$(1, 1, 1, 1)$ prior affects these estimates.

- *Third, to justify such broad claims about type I errors in clinical trial findings, this study should be much more rigid, with a predefined research question and a study protocol appropriately preregistered before the analysis is conducted. But the study is purely exploratory, and it remains unclear how the 71,126 CTGs were selected, why the described analysis pipeline ('We fit a ... model', etc.) is appropriate, and so on. On the positive side, I appreciate the availability of code on OSF.*

Our analysis was not prospectively preregistered, and we therefore make more cautious statements about Bayes/NHST contradictions in the revision. Our analysis is certainly exploratory, the goal being to reexamine published clinical trial results under a Bayesian lens when only limited information are available, and to find examples of the Jeffreys–Lindley paradox in clinical trails. For this purpose, eJAB is a useful methodology that can be applied easily.

We provide detailed inclusion and exclusion criteria for the clinical trials in Supplementary Material Appendix C, together with the sequential flow from 279,927 analyses in 22,454 studies to the final 16,563 analyses in 1,790 trials. Both significant and nonsignificant eligible results are retained. The appendix specifies the test mappings, sample-size and dimension rules, and treatment of reported inequalities. For left-censored values such as $p < .05$, we retain the inequality for significance classification and use the numerical bound for eJAB. This can overstate evidence for the null, so the resulting contradiction classifications are provisional.

- *Section 2: 'More generally, the volume of prior probability ellipsoids around the MLE is $\mathcal{O}(n^{\frac{1-q}{2}})$.' So what? Why is this particular prior choice being made? Is this good or bad? Without appropriate justification, the proposed prior and BF are (for $q > 1$) ad-hoc.*

The empirically derived prior distribution underlying eJAB is

$$\boldsymbol{\theta} \sim \boldsymbol{\mathcal{N}}_q\Big(\hat{\boldsymbol{\theta}},\ n^{\frac{1}{q}} \cdot \mathbf{J}_n^{-1}(\hat{\boldsymbol{\theta}})\Big)\,, \tag{8}$$

where $\mathbf{J}_n = \mathcal{O}(n)$ denotes the $q \times q$ information matrix. When $q = 1$, Eq. 8 reduces to the unit-information prior used in the BIC approximation and the original JAB (Aust et al., 2026; Wagenmakers, 2022). More generally, suppose that $\boldsymbol{\theta} \sim \boldsymbol{\mathcal{N}}_q\Big(\hat{\boldsymbol{\theta}},\ a_n \cdot \mathbf{J}_n^{-1}(\hat{\boldsymbol{\theta}})\Big)$, with $a_n > 1$. This yields $eJAB^*_{01} = a_n^{\frac{q}{2}} \exp\left\{-\frac{1}{2}\left(1 - \frac{1}{a_n}\right) Q_{\chi^2_q}(1-p)\right\}$ under the same approximation. Setting $a_n = n^{\frac{1}{q}}$ yields $\sqrt{n}$ growth in the evidence for $\mathcal{H}_0$, so that under eJAB the rate of evidence accumulation with sample size towards $\mathcal{H}_0$ is independent of the dimension q. When $q > 1$ (F- and chi-squared tests) and effect sizes are small, this avoids the overly conservative behavior of the BIC approximation to the Bayes factor in favor of the null hypothesis (as detailed in the `R` Markdown files in the GitHub Simulations directory), while retaining model-selection consistency under R1 and R2. To better demonstrate the performance of the resulting Bayes factor in comparisons with several alternative approaches (including TSBF) we evaluate the equally weighted decision error profiles.

Equally Weighted Decision Error Profiles: For each effect size θ and sample size n, we compare a simple null (effect zero) with a simple alternative (effect θ). Under 0–1 loss, the equally weighted decision error (EWDE) for decision rule d is

$$\mathrm{EWDE}_d(\theta, n) = \frac{\alpha_d(\theta, n) + \beta_d(\theta, n)}{2},$$

where α_d is the type I error probability and β_d is the type II error probability at the specified alternative. We define $\Delta E = \mathrm{EWDE}_{\text{comparison}} - \mathrm{EWDE}_{\text{eJAB}}$ and plot the normalized EWDE comparison,

$$\mathrm{EWDE}_{\text{norm}}(\theta, n) = \frac{\Delta E(\theta, n)}{M}, \qquad M = \max_{(\theta,n)\in\mathcal{G}} |\Delta E(\theta, n)|,$$

where $\mathcal{G}$ is the displayed grid for each panel. Positive values favor eJAB; negative values favor the comparison rule.

Figure 2 of the revised manuscript now displays $\mathrm{EWDE}_{\text{norm}}(\theta, n)$ and compares the performance of eJAB with: (A) the $p \leq .005$ decision rule, Benjamin et al. (2018); (B) the BIC approximation to the Bayes factor; (C) TSBF where the true standardized effect size is assumed to be known and the true

value is used in the Bayes factor function; (D) TSBF where the standardized effect size is consistently estimated as $\hat{\omega} = \sqrt{X^2/n}$, with X^2 the chi-squared statistic and n the sample size, and this estimate is used in the Bayes factor function. The BIC comparison uses $BF_{01}^{\text{BIC}} = n^{q/2}\exp(-\Lambda/2)$, where Λ is the likelihood-ratio statistic, for the six parametric-likelihood tests with available comparisons; Cox regression and the three rank tests are included for the comparisons with NHST but excluded for comparisons with BIC. The comparisons with TSBF focus strictly on chi-square tests for contingency tables. All Bayes factor approximations (eJAB, BIC, TSBF) use $BF_{01} < 1$ as the decision rule when evaluating error probabilities. Equality does not lead to rejection. NHST retains the rule $p \leq \alpha$. The standalone $p \leq .05$ comparison is provided in Supplementary Figure S3 (Appendix B); both NHST comparisons are also shown below. The manuscript abstract and discussion now have added discussion of these results.

First, with comparison to the $p <= 0.005$ decision rule (Figure 4), for all 10 settings considered we observe a region in the EWDE corresponding to effect sizes that are not too small or too large, where eJAB has mostly better performance. When comparisons are made to the $p <= 0.05$ decision rule (Figure 5), these same regions emerge when $q > 1$ for both parametric and nonparametric tests. The same can be said for the comparisons with the BIC approximation (Figure 6), where the EWDE exhibits a similar region of better performance for eJAB when $q > 1$. When $q = 1$ the performance of BIC and eJAB is close. The comparisons with TSBF show that TSBF will have equal or better performance than eJAB for chi-square tests when the Bayes factor function is evaluated at the true standardized effect size (Figure 7). In practice this parameter is unknown. When the standardized effect size is estimated, the situation changes, and eJAB becomes competitive, in particular for $q > 1$ and as q increases (Figure 8).

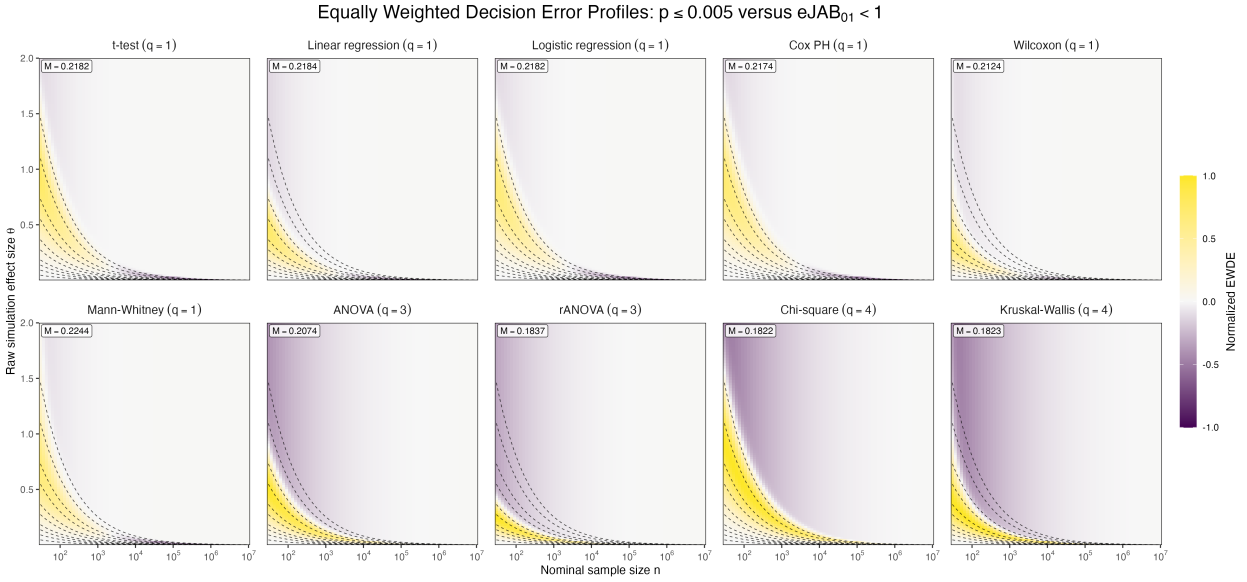


Figure 4: Equally Weighted Decision Error Profiles comparing eJAB with $p \leq .005$ across ten tests. The eJAB rule rejects $\mathcal{H}_0$ when $BF_{01} < 1$. Colors show normalized EWDE; positive values favor eJAB and negative values favor the NHST significance rule.

- *Section 3: A table with the different designs (the ten tests), with an additional column for the degree of freedom, would be helpful. Then the second paragraph in Section 3.1 can be shortened.*

We have added such a table and removed the second paragraph.

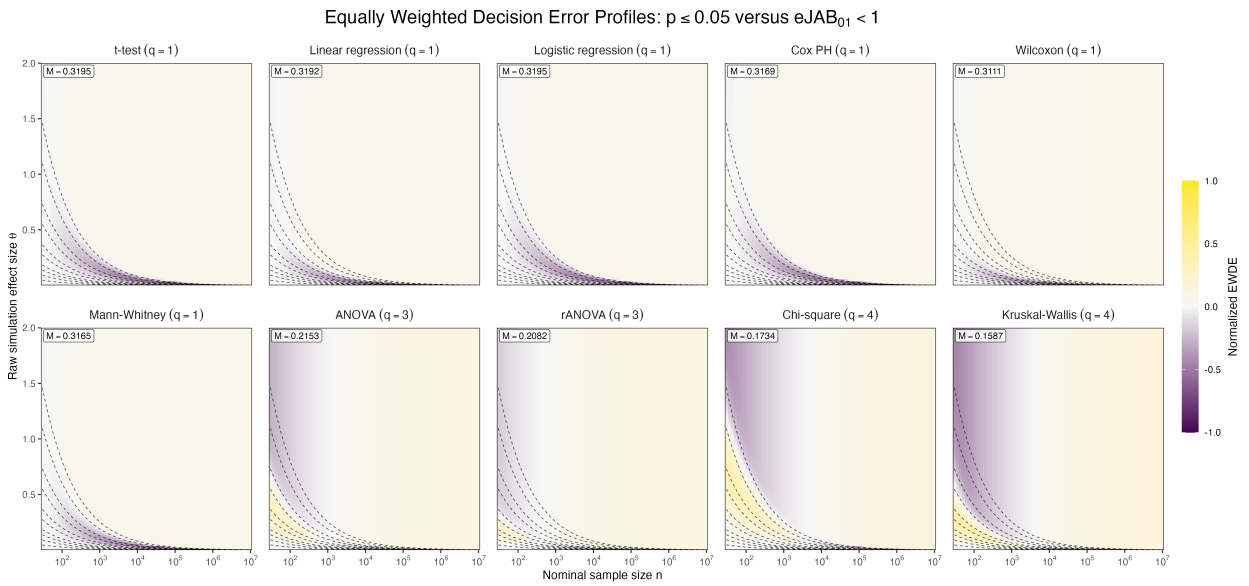


Figure 5: Equally Weighted Decision Error Profiles comparing eJAB with $p \leq .05$ across ten tests. The eJAB rule rejects $\mathcal{H}_0$ when $BF_{01} < 1$. Colors show normalized EWDE; positive values favor eJAB and negative values favor the NHST significance rule.

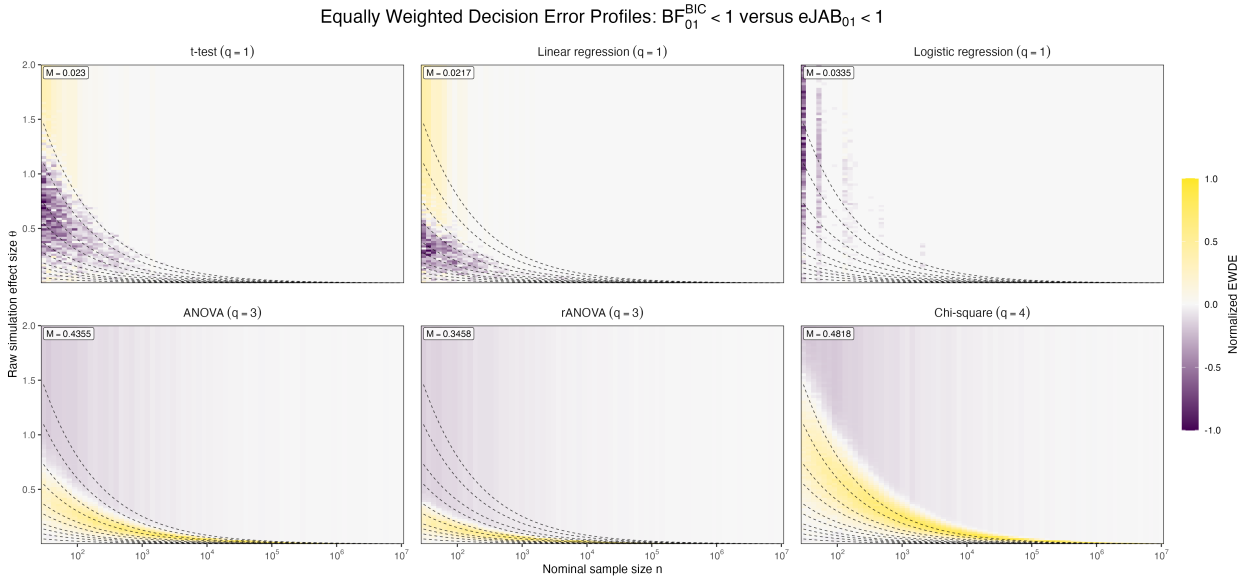


Figure 6: Equally Weighted Decision Error Profiles comparing eJAB with the BIC Bayes-factor rule across six parametric-likelihood tests: the two-sample t-test, linear regression, logistic regression, ANOVA, repeated-measures ANOVA, and chi-squared tests. Both Bayes-factor rules reject $\mathcal{H}_0$ when $BF_{01} < 1$. Colors show normalized EWDE; positive values favor eJAB and negative values favor BIC. Each panel reports its normalization constant M, the maximum absolute EWDE difference over the displayed grid.

- *Section 3: 'effective sample size for survival analysis': A reference to Volinsky and Raftery (2000) might be useful. Also, it is not clear to me how the survival and censoring times were sampled.*

We have added a reference to Volinsky and Raftery (2000) for the event-count definition of effective

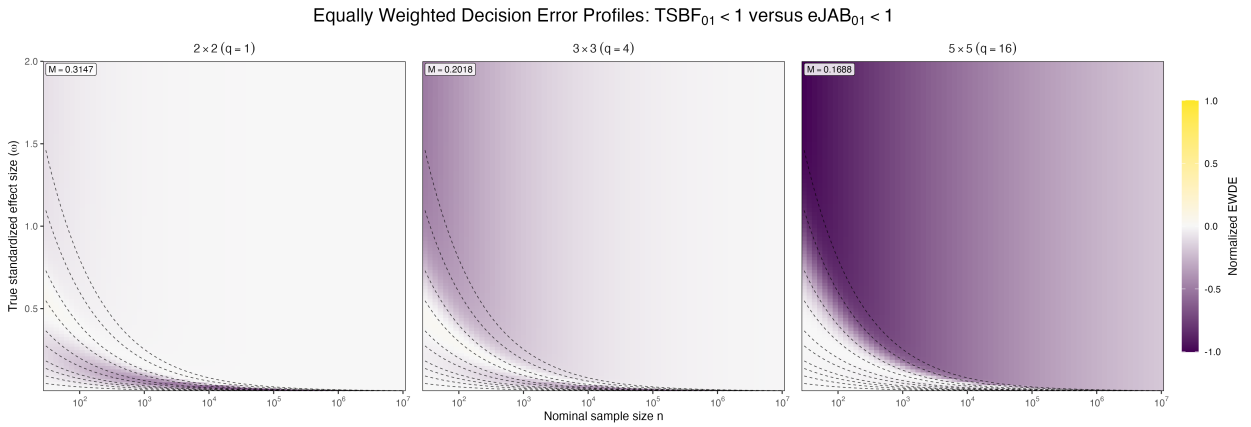


Figure 7: Equally Weighted Decision Error Profiles comparing eJAB with the TSBF for chi-square tests with 2×2, 3×3 and 5×5 tables, where the Bayes factor function is evaluated at the true standardized effect size. Colors show normalized EWDE; positive values favor eJAB and negative values favor BIC. Each panel reports its normalization constant M, the maximum absolute EWDE difference over the displayed grid.

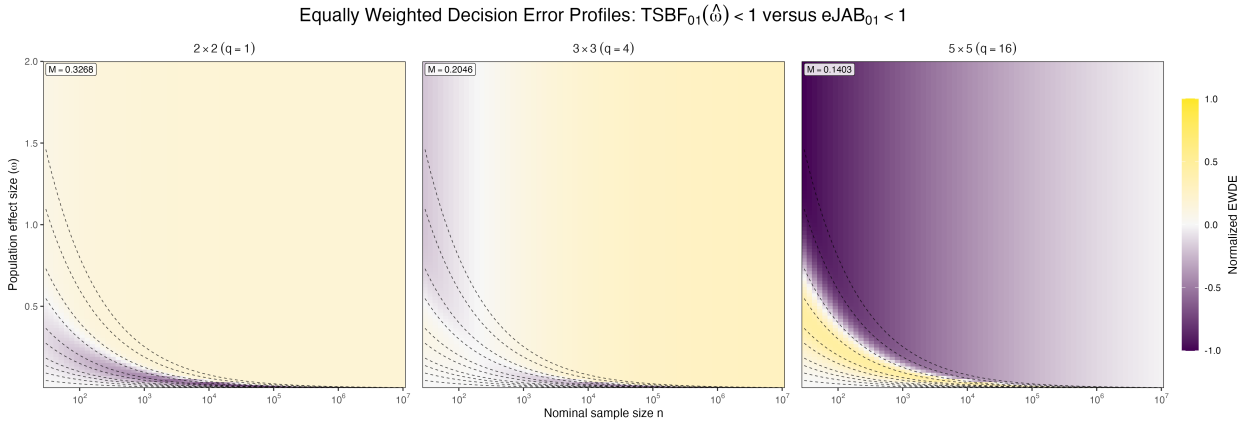


Figure 8: Equally Weighted Decision Error Profiles comparing eJAB with the TSBF for chi-square tests with 2×2, 3×3 and 5×5 tables, where the Bayes factor function is evaluated at the estimated standardized effect size. Colors show normalized EWDE; positive values favor eJAB and negative values favor BIC. Each panel reports its normalization constant M, the maximum absolute EWDE difference over the displayed grid.

sample size. The survival designs discussed here are simulation studies. Survival analyses are excluded from the revised CTG application when the required event counts are unavailable, as documented in Appendix C.

In the simulations, survival times and censoring times are generated independently for each subject. The treatment indicator is binary, with approximately equal numbers of subjects assigned to the two groups and the assignments randomly permuted. Conditional on treatment status x_i, the survival time is sampled from an exponential distribution, $T_i \sim \text{Exp}(\lambda e^{bx_i})$ for $i = 1, \cdots, n$, so that the baseline hazard is λ and the treatment hazard ratio is e^b.

Right-censoring is imposed in one of two ways. In the simulations using random censoring, the censoring time is sampled independently as $C_i \sim \text{Exp}(\lambda_c)$ independently of both the survival time and treatment assignment, corresponding to non-informative right censoring. Alternatively, when a maximum follow-up time is specified, all subjects are administratively censored at the fixed time $C_i = T_{\max}$. The observed follow-up time is $Y_i = \min(T_i, C_i)$, with event indicator $\delta_i = \mathbf{1}_{T_i \leq C_i}$. Thus, subjects are sampled independently, and the censoring mechanism is non-informative by construction.

We used the number of observed events, $n^* = \sum_{i=1}^{n} \delta_i$, as the effective sample size in the eJAB formula. The simulation code and accompanying description are provided in `coxphSim.Rmd` in the GitHub repository.

- *'In particular, we tested the harmonic mean of group sizes for item 1, as in JAB (Wagenmakers, 2022), but do not demonstrate it here': I assume this is intended to investigate unequal sample sizes in the two groups? Should this not then be half the harmonic mean?*

  The two sample-size considerations were not intended to address unequal group sizes per se. In the simulations, we used unequal group sizes with a ratio of 1.2:1, and we considered two sample-size conventions for computing Bayes factors: (1) the total number of observations and (2) one-half of the harmonic mean of the two group sizes.

  The former convention is commonly used in linear-regression formulations and in the BIC approximation. The latter is also the inverse-variance effective sample size, $1/n = 1/n_1 + 1/n_2$, and is used in the JAB and JZS Bayes factors (Morey and Rouder, 2026). In the `ttestSim.Rmd` supplementary file in the GitHub repository, we included both sample-size definitions when computing JAB and BIC.

  The use of one-half of the harmonic mean arises from the standard error of the difference between two independent group means. Wagenmakers (2022, pp. 26–27) discusses this sample-size choice in detail. Later, on p. 34, the central component is the $\sqrt{n}$ term—or, more generally, the sample-size quantity that enters the standard error of the maximum likelihood estimate.

- *Section 3.1: 'For local alternatives ...': Local alternatives only appear here, but they should also be discussed much more broadly in the theory section, as they form the basis of Johnson's BFs based on test-statistics.*

  We agree that local alternatives deserve a more prominent treatment in the theoretical development. We have expanded the discussion accordingly, including their role in TSBF and their relationship to our regularity conditions. These points are addressed in detail in the first bullet point of our response above.

- *Section 3.2: 'We compared eJAB with test-statistic BFs': How did you choose the parameter alpha (in Johnson's notation)? Or did you simply use the 'MMLE under the alternative hypothesis'? If so, it is not surprising that 'the test-statistic BF is inherently bounded below by 1'; it is actually a minimum BF! The discrepancies between your BF and Johnson's proposal are easy to explain, you are comparing apples with oranges.*

  We did indeed use the MMLE under the alternative hypothesis. In the revised manuscript, we instead compare the methods across several values of the hyperparameter and avoid comparisons based on the MMLE. These points are also addressed in detail in the first bullet point of our response above.

- *Beginning of Section 4: 'Prior work has raised concerns about the strength of statistical evidence in clinical research. For example, Begley and Ellis (2012) reported replication for only 11% of landmark preclinical cancer findings.' It does not make much sense to quote a study on the reproducibility of*

preclinical cancer findings as an example of 'concerns about the strength of statistical evidence in clinical research'. Clinical and preclinical research are very different fields.

Thank you for pointing this out. We have removed the reference from the revised manuscript.

- *The description of the 'candidate type I errors' is very repetitive, as you are always using the phrase 'We flag as a candidate type I error the reported effect ...' In addition, the reader would like to see additional information on sample size n (and dimension q) of the different studies, also because the focus is on the Jeffreys–Lindley paradox, a phenomenon that depends intrinsically on sample size. I noted that the first study is a four-arm Phase III trial, but only one p-value is reported. Is this the p-value for the comparison of all four groups or the p-value for the pairwise comparison of 50 mg vs placebo? What sample size was used, and did you also analyze the results from the other pairwise comparisons with different doses? If so, how did you deal with the dependence between the comparisons, given that they all share the same control arm?*

  In the revised manuscript, we report effect size, sample size, trial phase, and the outcome and comparison for each example. The three examples have $n = 66$, 80, and 1,586, respectively, with $q = 1$ in each case. The original four-arm Phase III trial is excluded by the revised eligibility criteria.

  More generally, a single study may contribute multiple p-values. For the posterior summaries, we combine one-sided and two-sided/unrestricted tests within studies into four categories defined by NHST significance ($p_1 \leq .025$ or $p \leq .05$, respectively) and whether the applicable eJAB exceeds 1. We use a Dirichlet$(1, 1, 1, 1)$ prior for each study and weight posterior category probabilities by the number of eligible analyses before calculating the contradiction proportion among significant results. By assigning each study its own category probabilities, the model allows for study-level heterogeneity.

- *End of Section 4: The linear model needs better motivation. Figure 4 would be more informative if the same y-axis scale were used throughout and if the panels were ordered A, C, B, D (for easier comparison of unadjusted and adjusted results).*

  Thank you for this suggestion. We have combined these results with the clinical-trial summaries in manuscript Figure 3. The outcome panels now appear as Figure 3D–G, with a common y-axis scale and unadjusted and adjusted results adjacent within each outcome type: D and E show primary outcomes, and F and G show secondary outcomes. This preserves the requested ordering of the former Figure 4 panels A, C, B, D.

  The linear mixed models describe within-study variation in $\ln eJAB$ and $\ln p$, separately for primary and secondary outcomes. We fit a fixed intercept and study-specific random intercept, and plot the residuals re-centered by the estimated fixed intercept.

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Response to Reviewer 2

There is no doubt that this work will attract a lot of attention. Similar to the 2018 Benjamin et al. paper, the approach outlined in this paper is straightforward to implement, and, in my opinion, there is a good argument to be made that when $p < \alpha$ ('reject $\mathcal{H}_0$') while $BF_{01} > 1$ ('the data support $\mathcal{H}_0$'), or even < 1 but $> 1/3$ ('the data provide weak evidence against $\mathcal{H}_0$'), this casts substantial doubt on the frequentist conclusion – at any rate, such a conflict demands a more careful analysis. The fact that such conflicts now go entirely unnoticed seems unproductive. It may be that, at several points throughout the paper, its accessibility can be increased even more. Regardless, the work provides a fresh perspective on a crucial issue that continues to haunt the field. It is certain to lead to a useful debate, and, in fact, I would suggest that –if it is accepted– it will be the target of a few comments.

Thank you very much for this encouraging and thoughtful assessment. We appreciate the reviewer's recognition of the practical relevance and potential impact of the proposed approach. We also agree that conflicts between NHST and Bayes-factor conclusions warrant closer scrutiny rather than going unnoticed (e.g., Aust et al., 2026; Tyner et al., 2025).

In the revised manuscript, we have expanded the theoretical content to provide a stronger justification for the methodology. We also agree on the importance of making the methodology more accessible. To this end, we have developed an eJAB calculator, available at `https://flxp0x-puneet-velidi.shinyapps.io/bayesfactor/`, and implemented eJAB in a JASP module designed to identify potentially irreplicable results. A companion article (in preparation) describes the JASP module and presents several examples.

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Alexander H. Tyner et al. Investigating the replicability of the social and behavioural sciences. *Nature*, 652: 143–151, 2025. doi: 10.1038/s41586-025-08258-3.